



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject : PROBABILITY & STATISTICAL DATA ANALYSIS
(SECI / SCSi 2143/ SECI 1143)
Task : Chapter 1 & Chapter 2

INSTRUCTION:

1. This is a **GROUP** assignment. Please clearly write the group members name & matric number in the front page of the submission.
 2. This assignment contributes to 5% of overall course marks.
 3. Only **HANDWRITTEN** submission is accepted:
 - a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc,) will be **REJECTED**.
 - b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
 - c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
 - d. Round your answers to **TWO** decimal places.
 - e. Please scan/snapshot your work and save as a PDF file.
 4. Submission via eLearning – only **ONE** group member needs to submit on behalf of the group.
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QUESTION 1[10 MARKS]

You are a research assistant at a university's Faculty of Computing. The faculty is conducting a study to explore the relationship between faculty students' socioeconomic backgrounds and their academic performance. The aim is to identify potential factors that may influence academic success and provide insights for educational policymakers and practitioners. Identify the following:

- a) What is population and sample?
- b) List possible variables/data to be collected and identify characteristic of the variables/data according to the following
 - i. Discrete/continuous
 - ii. Categorical/numerical

iii. Level of measurement: nominal, ordinal, ratio or interval

Give example of the sample of data to help your justification

- c) Primary Data Source
- d) Secondary data source
- e) Identify and construct possible descriptive statistics that can be used to summarize the data. Note : You only need to put any random figures in the constructed descriptive statistics.

QUESTION 2 [10 MARKS]

A student trainer worked in the training room in the physical education building. The number of students coming in the taped or for treatment of other injuries was recorded each half hour, yielding the following data. Calculate the frequency and relative frequency of the data

3	3	2	0	4	5	6	4	4	3
2	1	2	3	0	5	5	3	2	3
5	4	1	2	0	3	2	4	2	6

QUESTION 3 [25 MARKS]

A packaged product has a label weight of 450 grams. The company goal is to fill each packages with at least 450 grams, but at most 458 grams. To check these goals , a random sample of 100 packages was selected and weighted, yielding the following weights, rounded to the nearest gram:

450	450	458	460	450	472	455	462	463	467
453	455	466	453	471	451	472	464	463	453
469	457	460	458	472	454	473	466	455	470
470	472	452	459	453	466	457	464	462	459
472	450	471	464	453	472	457	461	450	470
471	467	456	465	450	450	457	465	466	454
472	467	455	456	455	470	471	471	450	458
469	467	457	455	463	453	458	452	456	469
457	472	460	459	471	461	470	453	469	471
450	454	451	450	456	472	465	464	471	467

Construct :

- a) Construct a histogram,
- b) frequency polygon
- c) ogive

QUESTION 4 [10 MARKS]

Below are the lifetime, in hours of fifty 40-watt 110-volt internally frosted incandescent lamps, taken from forced life tests :

919	1196	78	1126	936	918
1156	920	948	1067	1092	1162
1170	929	950	905	972	1035
1045	855	1195	1195	1340	1122
938	970	1237	956	1102	1157
978	832	1009	1157	1151	1009
765	958	902	1022	1333	811
1217	1085	896	958	1311	1037
702	923				

Construct a modified box-plot and determine the outliers (if any)